

Technical Specifications for Cross Examination

Overview

This document describes the simulation methods underlying Cross Examination (https://sruckman.shinyapps.io/cross_examination/), an R Shiny application for teaching genetic mapping. Source code is available at https://github.com/sruckman/Cross_Examination. The supplement is intended as a companion to the manuscript for readers who wish to understand or reproduce the simulation logic in full.

The app implements three tabs (MPP, Biparental QTL, and GWAS), each simulating a distinct study design. All three tabs share a common genetic map: a single chromosome represented as 202 evenly spaced positions spanning 100 cM, generated as `seq(0, 100, length.out = 202)` in R, giving a resolution of approximately 0.5 cM per position. The single-chromosome model isolates the core mapping logic from the bookkeeping of multi-chromosome genomes while preserving all relevant statistical properties: recombination frequency, linkage disequilibrium (LD) decay, and the relationship between sample size, effect size, and power. The key limitation is that multiple-testing correction in a real genome scales with the total number of markers across all chromosomes, not just the 202 positions used here. The fixed Bonferroni thresholds in the simulator are therefore specific to this single-chromosome representation and should not be applied directly to genome-wide data.

Teaching Mode is available on all three tabs as a checkbox in the sidebar. When enabled, it sets a fixed random seed (seed = 42) before simulation and fixes QTL positions to predetermined locations: 50 cM for a single QTL; 30 and 70 cM for two QTLs; 20, 50, and 80 cM for three QTLs; and 10 evenly spaced positions from 10 to 90 cM for the polygenic architecture. These positions correspond to marker indices 102, 62, 142, 42, 102, 162, and so on in the 202-position array. Fixing the seed and positions ensures that every student running identical parameter settings sees identical output, which is necessary for coordinated in-class exercises and for reproducing figures in teaching materials. When Teaching Mode is off, QTL positions are drawn randomly at each run using `sample()`, and results differ across runs.

Population Simulation

Recombination model

Crossovers are simulated under the Haldane mapping function (Haldane 1919), which assumes no crossover interference and models crossover events as a Poisson process along the chromosome. The Haldane function relates map distance d (in Morgans) to recombination frequency r as $r(d) = (1 - e^{-2d})/2$. At the 0.5 cM resolution of the simulator, adjacent marker positions have a recombination frequency of approximately $r = (1 - e^{-(0.01)})/2 = 0.005$.

In the simulator, the total number of crossover events per chromosome per generation is drawn from `Poisson(lambda)`, where `lambda` is set by design type as described below. Crossover positions are drawn from `Uniform(0, 100)` cM and sorted. Haplotype ancestry is initialized at position 0, then switched at each crossover breakpoint, so that the resulting chromosome is a mosaic of founder segments. After all crossovers are applied, each of the 202 marker positions is assigned the founder identity of the segment containing it.

Fully intercrossed design (MAGIC/DSPR-style)

Each simulated RIL is constructed by drawing an initial founder uniformly from all n_{founders} founders, then simulating n_{gen} rounds of crossover with `lambda = n_gen` (one expected crossover per 100 cM per generation). After each crossover, the new segment draws a founder uniformly from the full founder pool. This approximates the haplotype structure expected after n_{gen} generations of random intercrossing among n_{founders} , with each founder contributing equally, as in MAGIC (Kover et al. 2009) and DSPR (King, Merkes, et al. 2012; King, Macdonald, et al. 2012) populations.

Under this model, expected haplotype block length after n_{gen} generations is approximately $100/n_{\text{gen}}$ cM. For the default DSPR settings (8 founders, 50 generations), expected block length is $100/50 = 2$ cM, consistent with empirically estimated block lengths in the DSPR (King, Merkes, et al. 2012; King, Macdonald, et al. 2012). At 2 cM resolution, the 202-position chromosome supports approximately 50 distinct haplotype blocks per RIL, providing sufficient resolution to localize QTLs to within a few cM. Total individuals simulated per replicate equals $3 \times \text{pool_size}$. This three-pool structure (cases, controls, and unselected reserve) is described in detail below.

Hub-and-spoke design (NAM-style)

Each simulated RIL is assigned a hub founder (always founder 1) and one randomly chosen spoke founder drawn from founders 2 through n_{founders} . Crossovers are simulated with $\lambda = 0.1 \times n_{\text{gen}}$, and after each crossover, the segment switches between hub and spoke only. This reflects the two-founder ancestry structure of each individual RIL family in a Nested Association Mapping (NAM) design, in which each spoke line is crossed only to the common hub (McMullen et al. 2009).

Because each spoke allele is present in only approximately $1/(n_{\text{founders}} - 1)$ of all RILs, spoke-specific effect alleles are at lower frequency in the population than alleles in a fully intercrossed design, where each founder contributes to approximately $1/n_{\text{founders}}$ of all RILs. This difference in allele frequency affects the statistical power of BSA. To equalize expected signal-to-noise between the hub-and-spoke and fully intercrossed designs when effect sizes are comparable, effect sizes in the hub-and-spoke design are scaled by a correction factor hub_scale , computed as: $\text{hub_scale} = \sqrt{[f_{\text{magic}} \times (1 - f_{\text{magic}})]/[f_{\text{hub}} \times (1 - f_{\text{hub}})]}$, where $f_{\text{magic}} = 1/n_{\text{founders}}$ is the expected allele frequency per founder in the fully intercrossed design, and $f_{\text{hub}} = 0.5/(n_{\text{founders}} - 1)$ is the expected spoke allele frequency in the hub-and-spoke design (each spoke appears in half the RILs of one family, and there are $n_{\text{founders}} - 1$ families). This scaling factor is the ratio of the standard deviations of the frequency estimator under the two designs. The 0.1 multiplier on the crossover rate is a modeling simplification intended to reflect reduced effective recombination per position in a two-founder cross and is not derived from first principles.

Biparental F2

Each F2 individual is constructed by simulating two independent gametes from a heterozygous F1. Each gamete begins from a randomly chosen parent (P1 or P2, each with probability 0.5), and crossovers are drawn from $\text{Poisson}(1.0)$, reflecting one expected crossover per 100 cM of chromosome per meiosis, consistent with typical recombination rates in diploid organisms (Lander and Botstein 1989). After crossovers are applied, the gamete genotype at each of the 202 positions is P1 or P2. The two gametes are not combined into a diploid; instead, each gamete is treated as a haploid genotype (coded 1 or 2), which simplifies the regression model described below.

Since F2 individuals have experienced only a single generation of recombination from the F1, the expected haplotype block length is approximately $100 \text{ cM}/(\text{mean crossovers per gamete}) = 100/1 = 100$ cM. In practice, the $\text{Poisson}(1.0)$ distribution produces some individuals with zero crossovers (probability $e^{-1} = 0.37$) and some with several, giving a mixture of block lengths. This produces broad, poorly localized LOD peaks relative to multiparent designs with many generations of recombination, a contrast central to the simulator's teaching purpose.

Phenotype Simulation

Additive phenotype model

Phenotypes are generated under a purely additive model with Gaussian environmental noise. For individual i , the phenotype is: $y_i = \sum_j [\beta_j \times I(h_{ij} == \text{effect_founder}_j)] +$

$\epsilon_{1i} + \epsilon_{2i}$, where h_{ij} is the founder haplotype of individual i at QTL position j , β_j is the effect size assigned to the effect founder at locus j , $I()$ is the indicator function, $\epsilon_{1i} \sim N(0, 1)$ is a baseline noise term applied before QTL effects, and $\epsilon_{2i} \sim N(0, 1)$ is an additional noise term applied after QTL effects. Total residual environmental variance is therefore approximately 2. The ratio of total genetic variance (sum of per-locus variances) to total phenotypic variance determines the narrow-sense heritability of the simulated trait, which varies with architecture and effect sizes but is not explicitly reported by the simulator.

QTL model

A single QTL is placed at position index 102 (50 cM) in Teaching Mode or at a randomly drawn index in [10, 90] cM otherwise. The number of founders assigned effect alleles scales with n_{founders} (approximately 10% of available founders, minimum 1). Effect sizes are drawn from Uniform(1.125, 1.875), centered on 1.5. The effect size units are standard deviations of the environmental noise distribution, so a single effect allele with $\beta = 1.5$ shifts phenotype by $1.5/\sqrt{2} = 1.06$ environmental standard deviations.

2-QTL and 3-QTL models

Two or three QTLs are placed at fixed Teaching Mode positions or randomly drawn positions in [10, 90] cM. Effect sizes are drawn from Uniform(0.9, 1.5) for the 2-QTL model (centered on 1.2) and Uniform(0.75, 1.25) for the 3-QTL model (centered on 1.0). Per-locus effects are smaller to keep total genetic variance from growing excessively as locus number increases.

Polygenic model

Ten QTLs are placed at evenly spaced positions across [10, 90] cM in Teaching Mode (approximately every 8 cM) or at randomly drawn positions otherwise. Per-locus effect sizes are drawn from Uniform(0.525, 0.875), centered on 0.7. The total expected genetic variance across 10 loci is therefore roughly $10 \times 0.7^2 \times p \times (1 - p)$, where p is the frequency of the effect allele, giving a polygenic architecture with moderate aggregate heritability distributed across many loci of individually small effect. This architecture illustrates the challenge of detecting individual loci when genetic variance is distributed across many positions.

Biparental phenotype

For the F2, genotype at each position is coded as 1 (P1 haplotype) or 2 (P2 haplotype). The phenotypic contribution of a QTL at position j is: $\text{QTL contribution}_{ij} = (g_{ij} - 1.5) \times 2 \times \beta_j$, where g_{ij} is 1 or 2 and β_j is the assigned effect size. Subtracting 1.5 centers the contrast at zero, and multiplying by 2 ensures that the effect size parameter β_j has the same interpretation as in the MPP model: a 1-unit shift per effect allele copy. The heterozygote is not explicitly modeled because each gamete is treated as haploid; all individuals are effectively hemizygous at each position.

Case/control selection

After phenotypes are assigned, all $n_{\text{total}} = 3 \times \text{pool_size}$ individuals are ranked by phenotype score in descending order. The top pool_size individuals are assigned as cases. Controls are drawn by simple random sampling (without replacement) from the remaining $2 \times \text{pool_size}$ unselected individuals. The unselected pool of size $2 \times \text{pool_size}$ is retained to ensure that the control sample does not overlap with the cases and that there is a sufficient reserve from which to sample. This three-pool design follows the bulk segregant analysis framework described by Michelmores et al. (1991), in which extreme phenotype classes are compared to identify genomic regions where allele frequencies differ between groups.

Statistical Output

MPP - single replicate (G-test)

For each of the 202 marker positions, founder haplotype counts are tabulated separately in the case pool and the control pool. Since pool sequencing measures allele frequencies from pooled DNA rather than individual genotypes, observed allele counts are resampled from a multinomial distribution to simulate sequencing at a depth of 200 reads per pool. Specifically, if the case pool of size `pool_size` contains `n_f` individuals carrying founder `f` at position `p`, then observed read counts for the case pool are drawn as: `c_f ~ Multinomial(200, [n_1/pool_size, n_2/pool_size, ..., n_k/pool_size])` and similarly for the control pool. The resulting $2 \times k$ count table (rows: cases vs. controls; columns: `k` founder haplotypes) is used to compute the G-statistic: $G = 2 \times \sum_{ij} [\text{obs}_{ij} \times \ln(\text{obs}_{ij} / \text{exp}_{ij})]$, where `obsij` is the observed count for row `i` and column `j`, and `expij` = (row_{`i`}_total $\times$ col_{`j`}_total)/grand_total is the expected count under the null hypothesis of equal founder frequencies in cases and controls. The sum runs over all cells with `obsij` > 0. Degrees of freedom are $df = (2 - 1) \times (k - 1) = k - 1$. The G-test is asymptotically equivalent to the Pearson chi-square test under the null but performs better when expected cell counts are small (Sokal and Rohlf 1995). LOD is computed as: $\text{LOD} = -\log_p / \log(10)$, where $\log_p = \log(P(\text{chi-square}(df) \geq G))$ is computed on the log scale using R's `pchisq(G, df, lower.tail = FALSE, log.p = TRUE)` for numerical stability at very small p-values.

MPP - multiple replicates (CMH test)

When replicates > 1, each replicate `r` contributes an independent $2 \times k$ contingency table. Each replicate draws independent phenotype noise and independently simulates and selects pools from a freshly generated RIL population, so the tables are statistically independent across replicates. The Cochran-Mantel-Haenszel (CMH) test (Cochran 1954; Mantel and Haenszel 1959) combines these `R` independent tables into a single test of the common odds ratio across strata. The CMH statistic is: $\text{CMH} = [\sum_r (A_r - E_r)]^2 / \sum_r (V_r)$, where `Ar` is the observed count in the (cases, founder 1) cell of table `r`, `Er` = (row1_total_r $\times$ col1_total_r)/grand_total_r is its expected value under the null, and `Vr` is the hypergeometric variance of `Ar`: $V_r = (\text{row1_total_r} \times \text{row2_total_r} \times \text{col1_total_r} \times (\text{grand_total_r} - \text{col1_total_r})) / (\text{grand_total_r}^2 \times (\text{grand_total_r} - 1))$.

The CMH statistic is chi-square distributed with 1 degree of freedom under the null. This is implemented via R's `mantelhaen.test` with `correct = FALSE` (no continuity correction). LOD is computed from the CMH statistic, as in the G-test case.

The CMH accumulates signal at true QTL positions because the genetic effect is consistent across replicates (same QTL position, same founder carrying the effect allele), while noise at non-QTL positions is independent across replicates and cancels in the numerator sum. Magwene et al. (2011) demonstrated formally that replicated BSA analyzed with the CMH test provides substantially greater power than single-replicate designs for a fixed total number of individuals phenotyped.

Biparental - additive regression (F-test)

At each of 202 positions, an additive linear regression of phenotype on haplotype genotype (coded 1 or 2) is fit using R's `lm.fit(cbind(1, g), y)`, where `g` is the vector of genotype codes and `y` is the vector of phenotypes. The F-statistic for the additive effect is: $F = [(SS_{\text{null}} - SS_{\text{fit}}) / 1] / [SS_{\text{fit}} / (n - 2)]$, where $SS_{\text{null}} = \sum_i (y_i - \bar{y})^2$ is the total sum of squares under the null (intercept only), $SS_{\text{fit}} = \sum_i (r_i^2)$ is the residual sum of squares from the regression ($r_i = y_i - \hat{y}_i$), and `n` is the number of F2 individuals. The numerator has 1 degree of freedom (one predictor), and the denominator has `n - 2` degrees of freedom (`n` observations minus 2 parameters: intercept and slope). $\text{LOD} = -\log_{10}(p)$, where $p = P(F(1, n-2) \geq F_{\text{obs}})$. This is a simplified form of the regression interval mapping method of Haley and Knott (1992), without the flanking-marker interpolation step used in full interval mapping.

LOD smoothing

All LOD curves from the MPP and Biparental QTL tabs are smoothed with a 5-position rolling mean (spanning positions $i - 2$ through $i + 2$, approximately ± 1 cM) prior to plotting. Positions near the chromosome ends use a truncated window. This step reduces single-position noise introduced by the multinomial read-count resampling in the G-test/CMH computation, where stochastic variation in read sampling can produce isolated single-position spikes unrelated to QTL signal. Smoothing is applied only for visualization and does not affect the numerical LOD threshold or significance calls.

Significance thresholds

Bonferroni correction at family-wise error rate $\alpha = 0.05$ with $m = 202$ independent tests gives a per-test threshold of $\alpha / m = 0.05 / 202 = 2.475 \times 10^{-4}$, corresponding to $\text{LOD} = -\log_{10}(2.475 \times 10^{-4}) = 3.606$, rounded to 3.6 in the simulator. For the GWAS tab with $m = 1,000$ SNPs, the threshold is $0.05 / 1000 = 5 \times 10^{-5}$, corresponding to $\text{LOD} = -\log_{10}(5 \times 10^{-5}) = 4.301$, rounded to 4.3. These thresholds assume that the 202 marker positions (or 1,000 SNPs) represent statistically independent tests. In the MPP and biparental tabs, adjacent positions are correlated due to LD, so the Bonferroni threshold is conservative (the true significance threshold is somewhat lower). In the GWAS tab, SNPs within an LD block are perfectly correlated by construction, so the effective number of independent tests is $100 / B$ (number of blocks), not 1,000; the Bonferroni threshold is therefore highly conservative in the GWAS tab when LD blocks are large. Permutation-based thresholds (Churchill and Doerge 1994) are more accurate for correlated markers but are computationally expensive and not implemented in the simulator, which is designed for teaching rather than for rigorous significance testing.

GWAS Tab

Population model

The GWAS tab simulates an outbred population using a 12-founder haplotype mosaic model. Minor allele frequencies (MAFs) for each of the 1,000 SNPs are drawn independently from Uniform(0.05, 0.45). For each founder and each SNP, an allele (0 or 1) is drawn from Bernoulli(MAF_snp), producing a $12 \times 1,000$ founder haplotype matrix. Individual chromosomes are constructed by tracking which founder haplotype is active at each SNP position (the haplotype assignment vector $h1$ and $h2$ for the two chromosome copies), with haplotype switches between adjacent SNPs occurring independently at probability $p_{\text{switch}} = n_{\text{cross}} / (n_{\text{snp}} - 1)$, where $n_{\text{snp}} = 1,000$ and $n_{\text{cross}} = 100 / B$ for target average LD block size B cM. At each SNP position, each individual's genotype dosage is $d_i = \text{founder_hap}[h1_i, \text{snp}] + \text{founder_hap}[h2_i, \text{snp}]$, giving dosage 0, 1, or 2. To reduce memory usage at large sample sizes (up to 100,000 individuals), we retain only a subsample of 300 individuals' haplotype assignment vectors for visualization. The full genotype dosage matrix is used for the association scan.

LD block size parameter

For a target block size of B cM with 10 SNPs per cM, a block spans approximately $10 \times B$ SNPs. The expected number of haplotype switches per chromosome is $n_{\text{cross}} = 100/B$, and $p_{\text{switch}} = (100/B)/999$. At the default block size of 12 cM, $n_{\text{cross}} = 8.3$ and $p_{\text{switch}} = 0.0083$, giving an expected block of approximately 120 SNPs (12 cM $\times$ 10 SNPs/cM). Smaller block sizes (e.g., 5 cM) produce $p_{\text{switch}} = 0.020$ and expected blocks of 50 SNPs, representing populations with long recombination history such as diverse *Drosophila* or human African populations. Larger block sizes (e.g., 50 cM) produce $p_{\text{switch}} = 0.002$ and expected blocks of 500 SNPs, representing bottlenecked or domesticated populations.

GWAS scan

A chi-square test comparing allele dosage sums between cases and controls is performed at each of the 1,000 SNPs. Let C = sum of dosages in cases (maximum $2 \times n_{\text{case}}$), K = sum of dosages in controls (maximum $2 \times n_{\text{ctrl}}$), and $p_{\text{null}} = (C + K) / (2 \times (n_{\text{case}} + n_{\text{ctrl}}))$ be the pooled allele frequency estimate. Expected dosage sums under the null are $E_C = 2 \times n_{\text{case}} \times p_{\text{null}}$ and $E_K = 2 \times n_{\text{ctrl}} \times p_{\text{null}}$. The test statistic is: $\chi^2 = (C - E_C)^2/E_C + (K - E_K)^2/E_K$ with 1 degree of freedom. $\text{LOD} = -\log_{10}(P(\chi^2(1) \geq \text{observed } \chi^2))$. This is computed for all 1,000 SNPs simultaneously using vectorized operations in R. Cells where E_C or E_K is near zero (fixed SNPs) are assigned $\text{LOD} = 0$. Because all SNPs within a causal LD block share the same haplotype background by construction, all SNPs in the block show correlated allele frequency shifts between cases and controls and rise together in the Manhattan plot, illustrating the LD-driven broadening of association signals.

Case/control selection

Phenotypes are simulated for all n_{ind} individuals using the same additive plus Gaussian noise model described above. Cases are the top $n_{\text{ind}}/2$ individuals by phenotype; controls are a random sample of $n_{\text{ind}}/2$ from the remaining individuals.

Assumptions and Limitations

Single chromosome: All three tabs simulate a single 100 cM chromosome. Real mapping studies span multiple chromosomes, and genome-wide multiple-testing correction scales with the total number of markers. The 202-position and 1,000-SNP panels in the simulator correspond to a single chromosome, and the Bonferroni thresholds are calibrated accordingly.

Additive effects only: All QTL effects are purely additive. Dominance and epistasis are not modeled. Real quantitative traits frequently show non-additive genetic architecture (Mackay 2001), and the absence of dominance means the simulator does not capture the reduction in power that occurs when recessive alleles are rare or when dominance reversal occurs between environments.

No population structure or relatedness: Founders are assumed to contribute equally to the population, and no kinship matrix or genomic relatedness correction is applied. Real mapping populations, particularly natural populations used for GWAS, frequently show substantial stratification that can inflate chi-square statistics genome-wide and produce spurious associations (Price et al. 2006).

No genotyping or sequencing error: Genotype calls are perfect. Pool sequencing is approximated by multinomial resampling to a fixed read depth of 200 per pool, but allele calling error, contamination, GC bias, and mapping error are not modeled.

Fixed effect sizes: QTL effect sizes are assigned at simulation time and are not estimated from the data. The simulator does not produce point estimates, standard errors, or confidence intervals for effect sizes.

No selection during population construction: RIL populations are assumed to result from neutral intercrossing, with no viability selection, fertility selection, or inbreeding depression during population construction. Real multiparent populations may show segregation distortion that inflates allele frequency differences at specific loci.

Simplified GWAS LD model: The GWAS tab uses a discrete haplotype block model in which all SNPs within a block are in perfect LD ($r^2 = 1$ within a block, $r^2 = 0$ between blocks). Real LD decays continuously with distance and is imperfect even within defined blocks, described by statistics such as r^2 and D' (Lewontin 1964).

Gaussian environmental noise: Environmental effects are drawn from a normal distribution. Real trait distributions are often non-normal, and non-normality can affect the type I error rates of F-tests and chi-square tests.

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